

Machine Learning Concepts

Easy-to-understand study notes based on the PPT

Topics covered: Machine Learning types, Regression, Classification, Clustering, ML algorithms, Feature Engineering, Missing Values, Outliers, Encoding, Feature Scaling, Normalization, Standardization, Feature Selection and Correlation.

How to use these notes

Read the concepts in order. Focus on the simple idea first, then the example and finally the formula. The goal is to understand what a technique does, when it is used, and why it matters.

1. What is Machine Learning?

Machine Learning (ML) is a way of making computers learn patterns from data so they can make decisions or predictions without being explicitly programmed with every rule.

The PPT divides Machine Learning into four broad types:

- Supervised Learning
- Unsupervised Learning
- Semi-Supervised Learning
- Reinforcement Learning

A useful mental model is: data → learning process → useful output. The type of learning depends mainly on what information is available during learning.

2. Supervised Learning

In supervised learning, the data is labelled. We have input variables (X) and a known output/target (y). The model learns the relationship between them.

Basic idea: $X \rightarrow y$

Example: Age and salary are inputs, while Purchased (Yes/No) is the target.

The PPT divides supervised learning into two major categories: Regression and Classification.

3. Regression

Regression is used when the output is a numerical value.

- House price → ₹50 lakh
- Salary → ₹60,000
- Temperature → 32.5°C
- Sales → ₹1,50,000

Example: Given house area, number of bedrooms and location, a regression model can estimate the house price.

4. Classification

Classification is used when the output belongs to a category/class.

- Spam / Not Spam
- Yes / No
- Cat / Dog
- Pass / Fail

Example: An email can be classified as Spam or Not Spam.

5. Unsupervised Learning

In unsupervised learning, the data is unlabelled. We have input data (X), but there is no known target (y). The algorithm tries to discover patterns or groups by itself.

Example: Customer data containing age, income and spending score can be grouped into customer segments without giving the model the group labels.

The PPT connects this concept with clustering, including K-Means.

6. Semi-Supervised Learning

Semi-supervised learning uses a combination of a small amount of labelled data and a large amount of unlabelled data.

This can be useful when obtaining labels is expensive or time-consuming.

7. Reinforcement Learning

In reinforcement learning, an agent learns by taking actions in an environment and receiving rewards or penalties.

Basic flow: Agent → Action → Environment → Reward/Penalty → Learning

For example, a game-playing system can learn which actions lead to better outcomes.

8. Algorithms Mentioned in the PPT

Linear Regression: supervised regression; models a relationship between inputs and a numerical output.

Logistic Regression: used mainly for classification; it can produce a probability that is converted into a class.

Decision Tree: makes decisions through a sequence of learned questions/conditions, like a flowchart.

SVM (Support Vector Machine): finds a boundary that separates classes, with emphasis on the margin between them.

KNN (K-Nearest Neighbours): predicts using the labels/classes of nearby observations.

K-Means: unsupervised clustering algorithm that groups observations around centroids.

PCA (Principal Component Analysis): used for dimensionality reduction by transforming features into a smaller set of principal components.

9. Feature Engineering

A feature is an input variable/column used by a machine-learning model.

Feature Engineering means creating, transforming or modifying features so they are more useful for machine learning.

Examples: turning Date of Birth into Age, or combining Height and Weight into a derived measurement such as BMI.

The PPT focuses on practical preprocessing topics such as missing/null values, outliers, encoding, scaling and feature selection.

10. Null / Missing Values

A null or missing value means the value for an observation is unavailable.

Example: Salary = NaN for one row.

Common approaches include replacing numerical missing values with the mean or median, and using the mode or an 'Unknown' category for categorical data. The correct method depends on the dataset.

11. Outliers

An outlier is a value that is unusually far from most observations.

Example: 20, 22, 21, 23, 24, 200. The value 200 is an obvious unusual observation.

Outliers should be investigated rather than automatically deleted. They may represent errors, rare but real cases, or important events.

12. Encoding

Machine-learning algorithms generally need numerical input, but real-world data often contains categories such as Red, Green and Blue. Encoding converts categorical information into a numerical representation.

The PPT covers three methods: One-Hot Encoding, Label Encoding and Ordinal Encoding.

13. One-Hot Encoding

One-hot encoding creates a separate binary column for each category.

Example: Color = Red, Green, Blue

Red	Green	Blue
1	0	0
0	1	0
0	0	1

This avoids incorrectly suggesting that categories such as Red, Green and Blue have a numerical order.

14. Label Encoding

Label encoding assigns a numerical label to each category.

Example: Small → 0, Medium → 1, Large → 2.

The resulting numbers are category codes. Whether label encoding is appropriate depends on the nature of the categorical variable and the model.

15. Ordinal Encoding

Ordinal data has a meaningful natural order.

Example: Small < Medium < Large.

We can represent the ordered categories as Small → 0, Medium → 1, Large → 2 because the order itself carries information.

Encoding methods at a glance

Method	Use when	Example
One-Hot	Categories have no natural order	Red, Green, Blue
Label	Categories need numerical codes	Cat, Dog → codes
Ordinal	Categories have a natural order	Small, Medium, Large

16. Feature Scaling

Feature Scaling changes the numerical scale of features so that variables with very different ranges can be handled more appropriately by algorithms that are sensitive to magnitude or distance.

Example: Age may range from 20–60 while Salary may range from 20,000–200,000. Without scaling, the larger numerical scale can dominate distance calculations.

The PPT presents two main approaches: Normalization and Standardization.

17. Normalization (Min-Max Scaling)

Normalization in the PPT refers to Min-Max Scaling, which commonly maps values into the range 0 to 1.

Formula:

$$(X - X_{\min}) / (X_{\max} - X_{\min})$$

Example: If $X = 2$, minimum = 0 and maximum = 15:

$$(2 - 0) / (15 - 0) = 2/15 \approx 0.13$$

So the original value 2 becomes approximately 0.13 after scaling.

18. Standardization

Standardization transforms values so that the resulting variable has approximately mean 0 and standard deviation 1.

Formula:

$$Z = (X - \mu) / \sigma$$

where X is the original value, μ is the mean and σ is the standard deviation.

PPT example: Data = 2, 3, 1, 6; mean = 3; SD ≈ 1.85 .

For $X = 2$: $Z \approx -0.54$. For $X = 3$: $Z = 0$. For $X = 1$: $Z \approx -1.08$. For $X = 6$: $Z \approx 1.62$.

19. Normalization vs Standardization

Normalization: commonly maps values to a fixed range such as 0–1.

Standardization: centers the data around mean 0 and scales it to standard deviation 1.

Memory trick: Normalization $\rightarrow$ fixed range. Standardization $\rightarrow$ mean 0, SD 1.

20. Feature Selection

Feature Selection means selecting useful features and removing unnecessary or redundant ones.

Example: if a dataset contains Age, Salary, Experience, Height, Weight and an ID column, an ID may be irrelevant to the learning task and can be excluded.

Why select features? Too many irrelevant features can increase complexity, computation and noise, and can make models harder to interpret.

21. Correlation

Correlation measures the strength and direction of a linear relationship between two variables.

The correlation coefficient generally ranges from -1 to $+1$.

- Positive correlation: as X increases, Y tends to increase.
- Negative correlation: as X increases, Y tends to decrease.
- Near zero correlation: little or no linear relationship.

Example: A correlation of 0.89 indicates a strong positive linear relationship.

22. Correlation for Feature Selection

If two features have an extremely high correlation, they may contain very similar information. One may sometimes be removed to reduce redundancy.

Example: Age in years and Age in months are almost perfectly correlated, so keeping both usually adds little new information.

Important: correlation measures linear association. A low correlation does not automatically mean a feature is useless, because useful nonlinear relationships can exist.

23. K-Means in Context

K-Means is the unsupervised clustering algorithm highlighted in the PPT. It groups observations by their distance from cluster centroids.

Basic K-Means flow:

- Choose K , the number of clusters.
- Initialize K centroids.
- Calculate distance from each point to each centroid.
- Assign each point to its nearest centroid.
- Recalculate each centroid using the mean of its cluster.
- Repeat until the assignments/centroids stabilize or the stopping condition is reached.

Key idea: K-Means = find centers $\rightarrow$ measure distance $\rightarrow$ assign points $\rightarrow$ move centers $\rightarrow$ repeat.

24. Quick Revision Sheet

Machine Learning: learning patterns from data.

Supervised: labelled data; input X and target y .

Regression: numerical output.

Classification: categorical output.

Unsupervised: unlabelled data; discover patterns/groups.

Semi-Supervised: small labelled + large unlabelled data.

Reinforcement Learning: actions + rewards/penalties.

Feature: input variable/column.

Feature Engineering: create/transform features for ML.

Outlier: unusually distant observation.

Null: missing value.

One-Hot: separate binary column for each category.

Label Encoding: numerical code for categories.

Ordinal Encoding: numerical representation for ordered categories.

Normalization: commonly scales to a fixed range such as 0–1.

Standardization: mean 0, standard deviation 1.

Feature Selection: keep useful features, remove unnecessary/redundant ones.

Correlation: strength and direction of linear relationship, from -1 to $+1$.

K-Means: unsupervised algorithm that groups points around centroids.

25. The Big Picture

A practical way to connect the PPT concepts is:

Raw Data → Understand Features → Handle Missing Values/Outliers → Encode Categories → Scale Features when appropriate → Select Useful Features → Train an ML Algorithm → Evaluate/Use the Result

Different algorithms require different preprocessing. For example, distance-based methods such as K-Means are especially sensitive to feature scale, while some tree-based methods are much less sensitive to scaling.

26. Most Important Things to Remember

- First identify whether the problem is supervised or unsupervised.
- For supervised learning, decide whether the target is numerical (regression) or categorical (classification).
- For unsupervised learning, clustering can discover groups without target labels.
- Categorical variables may need encoding.
- Scaling matters when feature magnitudes or distances influence the algorithm.
- Feature selection removes unnecessary or redundant inputs.

- Correlation tells you about linear association, not causation.
- For K-Means, understand centroids, Euclidean distance, cluster assignment and repeated centroid updates.